

Apache AsterixDB

Scalable Big Data Management for Geospatial Analytics

- Open source - Apache 2.0 Licensed with an active community involving academia and industry
- Parallel Processing with shared-nothing architecture, horizontally scalable
- Flexible Data Model - Based on JSON, truly NoSQL
- SQL++ Query Language - Declarative queries with SQL like Syntax



Geospatial Capabilities

1. Spatial data treated as a first-class citizen
 - a. Unified Queries - Combine spatial predicates and non-spatial filters, no ETL required
2. OGC Simple Features Compliant
3. 40+ Spatial Functions including functions for
 - a. geometry construction -: `st_make_point`, `st_geom_from_geojson`, `st_geom_from_text`
 - b. Spatial Predicates -: `st_intersects`, `st_contains`, `st_within`, `st_distance`
 - c. Spatial analysis -: `st_union`, `st_intersection`, `st_buffer`, `st_area`
4. Distributed R-tree index integrated with LSM storage engine
 - a. Accelerates range queries and spatial joins at scale
5. Optimized Spatial Joins for parallelism
 - a. Implementation of Partition-Based Spatial-Merge algorithm with plane-sweep
 - b. Index-free parallel join for large-scale spatial analytics

Current Developments

1. Coordinate Reference Systems (CRS) Support with Apache SIS
2. Support for Shapefile, GeoParquet with projection and filter pushdown
3. Top-K Spatial Query support
4. Expanding Function Library - Additional OGC functions & spatial aggregations

Join the sprint! We're a friendly, inclusive team & welcome contributions in all forms :)

Get Started at ,

asterixdb.apache.org

github.com/apache/asterixdb